

Project Design Phase-II

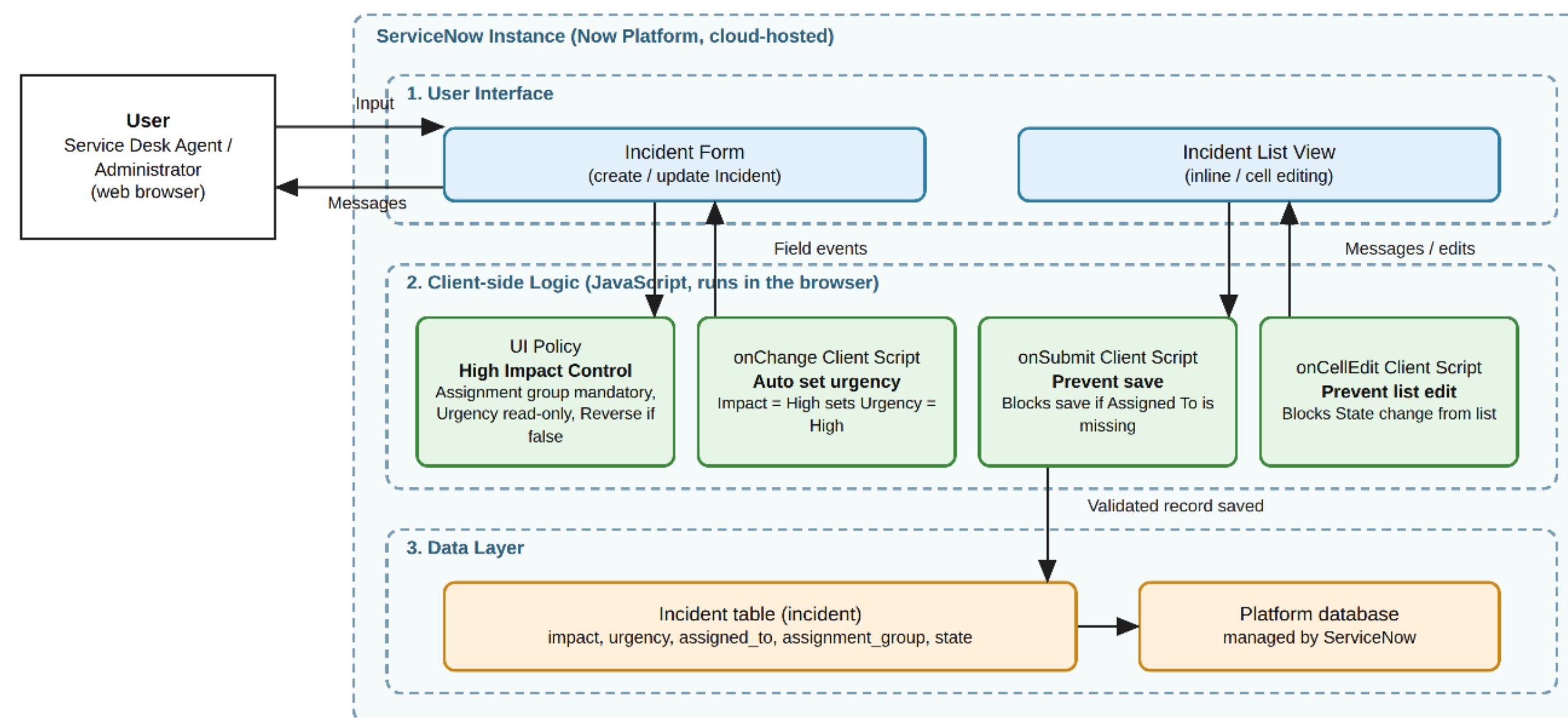
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	9100IncidentGuard
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Architecture: UI Policy and Client Scripts validate Incident data on the ServiceNow form and list view before saving.



Not used in this solution: external APIs, machine learning model, separate file storage.

Step	Data flow in the architecture
1	The administrator logs in to the ServiceNow instance with configuration access and opens System UI to create the rules.
2	A UI Policy 'High Impact Control' is created on the Incident table: when Impact is High, Assignment group becomes mandatory and Urgency becomes read-only (Reverse if false enabled).
3	An onChange client script 'Auto set urgency for high impact' runs when Impact changes and sets Urgency to High with an information message.
4	An onSubmit client script 'Prevent save if Assigned To missing' runs when the form is saved and blocks the save with an error message if Impact is High and Assigned To is empty.
5	An onCellEdit client script 'Prevent state change via list edit' blocks State changes made directly in the Incident list view and shows an alert.
6	Valid Incident records are saved to the Incident table in the ServiceNow database; testing confirms the mandatory check, successful save, reverse condition, list edit blocking and form-based update.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Incident form and Incident list view of the ServiceNow platform, used by agents to create and update Incident records.	ServiceNow Platform UI (HTML, CSS, JavaScript)
2.	Application Logic-1	UI Policy 'High Impact Control' on Incident: when Impact is High, Assignment group becomes mandatory and Urgency becomes read-only (Reverse if false).	ServiceNow UI Policy and UI Policy Actions
3.	Application Logic-2	onChange client script 'Auto set urgency for high impact' and onSubmit client script 'Prevent save if Assigned To missing'.	JavaScript (ServiceNow Client Script API, g_form)

4.	Application Logic-3	onCellEdit client script 'Prevent state change via list edit' blocks State changes made directly in the list view.	JavaScript (ServiceNow Client Script API)
5.	Database	Incident table storing the validated records (Impact, Urgency, Assigned To, Assignment group, State).	ServiceNow relational database (platform managed)
6.	Cloud Database	Records are stored in the database of the cloud-hosted ServiceNow instance; no separate database service is set up.	ServiceNow Now Platform (cloud)
7.	File Storage	Not required by the solution. Project documents and screenshots are kept in shared project folders.	N/A
8.	External API-1	Not used. All logic runs inside the ServiceNow instance.	N/A
9.	External API-2	Not used.	N/A
10.	Machine Learning Model	Not used.	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on the cloud-hosted ServiceNow instance. Local Server Configuration: not applicable. Cloud Server Configuration: UI Policy and Client Scripts created in the instance through System UI with administrative access.	ServiceNow cloud instance (Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is added; the solution uses the built-in client-side scripting of ServiceNow.	JavaScript (ServiceNow Client Script API)
2.	Security Implementations	Configured only by users with administrative access. Save-time validation blocks incomplete High impact	ServiceNow roles and access controls, UI Policy, onSubmit and onCellEdit scripts

S.No	Characteristics	Description	Technology
		Incidents and list-edit blocking protects State. These client-side rules complement server-side platform security.	
3.	Scalable Architecture	Runs on the multi-tier ServiceNow platform (browser, application server, database); the pattern can be reused for other fields or tables.	ServiceNow Now Platform
4.	Availability	Availability depends on the hosted ServiceNow instance; the rules are part of the instance, so no extra servers are needed.	ServiceNow cloud hosting
5.	Performance	Scripts run in the browser without a server round trip; onChange exits early when the form is loading or the value is empty.	Client-side JavaScript (g_form API)